

UTSAV AMRISH JOSHI

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Summary

- Mechanical engineer with plant floor experience in time studies, bottleneck analysis, and equipment performance validation.
- Built capacity and ROI models that shaped automation and expansion decisions at a high-volume food processing plant.
- Diagnosed mechanical, thermal, and electrical failure modes on industrial equipment through staged load testing and inspection.
- Comfortable running the data collection myself, then turning it into a documented recommendation someone can act on.

Education

Master of Science, Mechanical Engineering

Arizona State University

08/2024 - 05/2026

Tempe, Arizona

Bachelor of Engineering, Mechanical Engineering

Gujarat Technological University

08/2020 - 06/2023

Ahmedabad, India

Skills

Tools: Vibration & Alignment Testing, FAT/PDI Protocols, Pump Sizing

Production: Time Study & Bottleneck Analysis, Throughput Modeling, Capacity Planning, ROI/Payback Analysis, Process Mapping

Quality: Tolerance Stack-up, GD&T, SQF, FDA/USDA Compliance, Audit Readiness, Lean Six Sigma, Root Cause Analysis, SPC

Software: MS Office (Excel, Outlook, Word, PowerPoint, Teams), SolidWorks, AutoCAD, Siemens NX, ANSYS, COMSOL

Certifications: CAD Certification (SolidWorks, AutoCAD, Siemens NX); Lean Six Sigma Yellow Belt - ASU

Work Experience

Manufacturing Engineering Intern | Shamrock Foods | Phoenix, Arizona, USA

05/2026 - 07/2026

- Projected **25%** capacity growth for a plant expansion via conceptual automated handling system design.
- Redesigned process equipment layout using AutoCAD to accommodate future projects and equipment.
- Built a capacity tracker consolidating equipment specs & throughput ratings via time studies & bottleneck analysis.
- Integrated an AI documentation retrieval tool (Copilot Studio & SharePoint) for the QA department.
- Improved audit readiness by validating an AI documentation system across **65 docs** and **7+ compliance** domains.
- Completed OSHA safety orientation and ammonia/PSM refrigeration training; applied PPE protocols on-site.

Manufacturing Engineer | Gangadhar Industries | Vapi, Gujarat, India

07/2023 - 04/2024

- Reduced rework risk on **5-625 kVA gensets** by validating alignment, vibration, temperature, & electrical stability.
- Diagnosed alignment and vibration issues using dial indicators, laser alignment tools & vibration meters.
- Secured shipment readiness via Pre-Delivery Inspections, verifying enclosure & anti-vibration mount performance.
- Logged thermal, vibration & electrical data for 6 gensets in FAT checklists, creating a dispatch record.

Mechanical Engineer Intern | Linde Engineering | Vadodara, Gujarat, India

02/2023 - 04/2023

- Extracted static head and piping data from P&IDs across two process plants to define pump duty points.
- Calculated TDH and Darcy-Weisbach friction losses to quantify total system resistance per duty point.
- Verified NPSHa $\geq$ NPSHr and efficiency margins across **3-5 pump** options per duty point via design reviews.
- Sized high-head multistage pumps by factoring heat-exchanger drop and high-temp fluid effects into TDH/NPSH.

Mechanical Quality Intern | Chamunda Engineering Works | Bhavnagar, Gujarat, India

05/2022 - 08/2022

- Enforced quality standards on machined Bell Crank levers via **10% acceptance** sampling of daily lots.
- Verified dimensional and bore accuracy using GO/NO-GO and bore gauges to catch out-of-spec parts.
- Maintained daily lot traceability by segregating OK, rework & nonconforming parts via visual inspection and logs.
- Identified burrs and broken-edge defects across multistage CNC machining by collaborating with operators.

Projects

Coupled Electromagnetic - Thermal Modeling of Microwave Heating | COMSOL | ASU

01/2026 - 08/2026

- Built a coupled Maxwell-heat-transfer model in COMSOL (**2.45 GHz cavity, 250 mL water load**) for microwave heating.
- Validated hand-calculated penetration depth & energy balance; modeled permittivity via a **Debye relaxation** model.

Transient Thermal Simulation - Additive Manufacturing | ANSYS, COMSOL | ASU

01/2025 - 04/2025

- Modeled FDM layer-by-layer heat buildup in ANSYS (element activation) comparing ABS and PA12 heat-affected zones.
- Modeled laser powder-bed heating in COMSOL (Ti-6Al-4V); circular scans peaked at **~2405 K** vs **~2264 K** linear.

Graduate Teaching Assistant | Arizona State University | MEE 351, MAE 545

08/2025 - 05/2026

- Communicated technical concepts via in-class COMSOL, ANSYS, and MATLAB demos, grading **100+** students.
- Redesigned quiz and grading format for consistent, equitable outcomes across **170** students.